

OMI Gauge Table, F-Column Surface, and Byte-Ring Closure

Status: Canonical working draft

Layer: notation / gauge table / proof bridge / port surface

Scope: OMI workbook notation, OMI-Port carrier notation, OMI-ISA witness fixtures, Coq proof alignment

1. Core Claim

OMI notation begins in the hidden control plane, crosses into readable separation at `0x20`, branches through the printable F-column surface, and closes through the Null Ring byte cycle.

The system has four linked surfaces:

Pinch Point	0x00..0x1F
Gauge Control	0x1C..0x1F
Space Boundary	0x20
Branch Surface	0x1F..0x6F
Gauge Table	0x00..0x7F
Surrogate Field	0x7F..0xFF

The hidden control plane is:

0x1C	FS	File / Frame Separator
0x1D	GS	Group / Graph Separator
0x1E	RS	Record / Relation Separator
0x1F	US	Unit / Symbol Separator

The printable F-column surface is:

0x1F	US	hidden unit separator
0x2F	/	printable separator
0x3F	?	witness/query separator
0x4F	0	upper/norm Omicron
0x5F	_	carrier/floor separator
0x6F	o	lower/local omicron

Canonical short rule:

FS GS RS US are the hidden control gauges.
US / ? 0 _ o is the printable F-column branch.

2. Hidden Gauge Control and Tangential Projection

The non-printing control plane runs from:

0x00..0x1F

It ends in the gauge quartet:

0x1C	FS
0x1D	GS
0x1E	RS
0x1F	US

This quartet is the tangential gauge operator for the system.

It is linear in the byte table:

0x00 ... 0x1C 0x1D 0x1E 0x1F

But when viewed through the ASCII table as a two-dimensional projection of a one-dimensional byte line, the hidden gauge quartet becomes the branch base for the printable F-column:

0x1F	US
0x2F	/
0x3F	?
0x4F	0
0x5F	_
0x6F	o

Thus:

0x1F is the hidden unit separator.
0x2F is the first printable path separator.
0x3F is the query/witness separator.
0x4F is upper Omicron.

```
0x5F is the carrier floor.  
0x6F is lower omicron.
```

The space byte is the hinge:

```
0x20  SP
```

At `0x20`, the hidden control plane becomes readable separation.

3. Space, Branch, Gauge Table, and Surrogates

The system may be read as a staged byte surface:

```
Pinch Point:  
  0x00..0x1F  
  
Gauge Control:  
  0x1C..0x1F  
  
Space:  
  0x20  
  
Branch Point:  
  0x1F..0x6F  
  
Gauge Table:  
  0x00..0x7F  
  
Interpretive Space:  
  0x80  
  
Complementary Surrogate Annotations:  
  0x7F..0xFF
```

The first readable separation occurs at:

```
0x20
```

The first deletion/seal boundary occurs at:

0x7F

The saturated carrier horizon occurs at:

0xFF

The interpretive midpoint above the seven-bit gauge table is:

0x80

The local/private/public/remote interpretation of 0x80..0xFF remains an open declarative-resolution layer. It should not be overclaimed yet.

Working placeholder:

```
0x80..0x8F  declared frame constructions
0x90..0x9F  buffer configurations
0xA0..0xAF  attestation and scribe-facing annotations
0xB0..0xBF  escape / alias / carrier transition space
0xC0..0xFF  place-value extension and surrogate annotation space
```

Boundary:

The upper byte field may annotate, surrogate, escape, or extend.
It does not automatically validate or receipt.

4. Omicron Resolver Envelope

The Omicron resolver envelope is parameterized by an F-band gauge:

Fδ 00 1C 1D 1E 1F 20 Fδ

where:

$F\delta \in 0xF0..0xFF$

The canonical closure envelope is:

```
FF 00 1C 1D 1E 1F 20 FF
```

Meaning:

```
F8  selected F-band gauge / resolver horizon
00  NUL origin
1C  FS frame boundary
1D  GS group boundary
1E  RS relation boundary / diagonal closure witness
1F  US unit boundary
20  readable space boundary
F8  matching resolver closure
```

Authority boundary:

```
Omicron frames.
Omicron does not validate.
Omicron does not receipt.
```

5. OMI Notation Citation Surface

OMI notation may be written as a citation ket:

```
|OMI---IMO>
```

or as the folded route:

```
omi---imo
```

The transport carrier begins with:

```
omi---imo US/
```

where `US/` means:

```
0x1F  hidden unit separator
0x2F  printable separator
```

The printable branch alias is:

```
?0_o
```

Therefore the compact canonical alias is:

```
omi---imo?0_o
```

This may be read as:

```
omi---imo
  root folded relation

?
  witness/query separator

0
  upper/norm Omicron

-
  carrier/floor separator

o
  lower/local omicron
```

Expanded relation surface:

```
omi---imo/?---?0---0_---_o---o
```

Canonical distinction:

```
#/ belongs to carrier compatibility syntax.
?0_o belongs to OMI-native notation.
```

6. Place-Value Cascade

The branch surface cascades through hexadecimal place-value lanes.

The compact OMI surface may begin with:

```
omi---imo US/
```

and cascade through `?0_o` as the visible place-value branch:

```
(0x0000, 0x1000, 0x2000, 0x3000)
(0x4000, 0x5000, 0x6000, 0x7000)
(0x8000, 0x9000, 0xA000, 0xB000)
(0xC000, 0xD000, 0xE000, 0xF000)
```

This is the high-nibble lane surface.

Interpretation:

```
0x0000..0x3000  first quadrant / origin-facing route
0x4000..0x7000  second quadrant / printable branch route
0x8000..0xB000  interpretive/surrogate route
0xC000..0xF000  extension/escape/place-value route
```

The precise public/private/local/remote semantics of the upper quadrants should remain declarative until they are proven or implemented.

7. Null Ring and Byte-Ring Closure

The folded null seed is:

```
(NULL . NULL)
```

The byte ring is:

```
(0x00 . 0x20)
(0x20 . 0x7F)
```

```
(0x7F . 0xFF)
(0xFF . 0x00)
```

The dot relation is XOR:

```
0x00 ^ 0x20 = 0x20
0x20 ^ 0x7F = 0x5F
0x7F ^ 0xFF = 0x80
0xFF ^ 0x00 = 0xFF
```

The full witness closure is:

```
0x20 ^ 0x5F ^ 0x80 ^ 0xFF = 0x00
```

The four absences are:

```
0x00  absence before inscription
0x20  absence inside readable text
0x7F  absence after removal
0xFF  saturated absence
```

Canonical law:

```
The Null Ring closes because every boundary appears twice in the full cycle.
```

Expanded cancellation:

```
0x00 ^ 0x20 ^ 0x20 ^ 0x7F ^ 0x7F ^ 0xFF ^ 0xFF ^ 0x00 = 0x00
```

8. Metatron Witness

Metatron is the attestation scribe and contrast witness.

Canonical registers:

```
0xAA55
0x55AA
```


Binary form:

```
0xAA55 = 10101010 01010101
0x55AA = 01010101 10101010
```

Metatron laws:

```
0xAA55 has 8 one bits and 8 zero bits.
0x55AA has 8 one bits and 8 zero bits.
0xAA55 ^ 0x55AA = 0xFFFF.
0xAA55 ^ 0xFFFF = 0x55AA.
0x55AA ^ 0xFFFF = 0xAA55.
```

Boundary:

```
Metatron scribes.
Metatron witnesses.
Metatron interpolates.
Metatron does not validate.
Metatron does not receipt.
```

9. Tetragrammatron Diagonal Closure

The Tetragrammatron adjudicates closure through the Polybius diagonal witness.

The nibble field is:

```
0  1  2  3
4  5  6  7
8  9  A  B
C  D  E  F
```

The two diagonals are:

```
D+ = {0, 5, A, F}
D- = {3, 6, 9, C}
```

XOR closure:

```
0 ^ 5 ^ A ^ F = 0x0
3 ^ 6 ^ 9 ^ C = 0x0
```

The full diagonal set is:

```
D+ ∪ D- = {0, 3, 5, 6, 9, A, C, F}
```

The complement is:

```
K = {1, 2, 4, 7, 8, B, D, E}
```

Sum invariants:

```
SUM(D+) = 0x1E
SUM(D-) = 0x1E
SUM(D+ ∪ D-) = 0x3C
SUM(K) = 0x3C
SUM(0..F) = 0x78
```

Thus:

```
0x1E  relation closure witness
0x3C  centered fold witness
0x78  shifted/full system witness
```

Tetragrammatron fold laws:

```
0x3C is the centered fold witness.
0x78 is the shifted/carry fold witness.
0x78 == (0x3C << 1) masked to 8 bits.
0x3C == (0x78 >> 1).
0x3C has 4 one bits and 4 zero bits.
0x78 has 4 one bits and 4 zero bits.
0x3C ^ 0x78 == 0x44.
```

Boundary:

Tetragrammatron validates and fold-carries.
It does not project.
It does not scribe.

10. Tetragrammatron Polyharmonic Governor

The Tetragrammatron is also the Polyharmonic Governor.

It separates:

3 clocks
4 visible offsets
5 governor modes

10.1 Three Clocks

Atomic Logic Clock
Spectral Observer Clock
Cosmic Orbit Clock

10.2 Four Visible Offsets

FS = 0x0001
GS = 0x0010
RS = 0x0100
US = 0x1000

Offset stepping:

0x0001 << 4 == 0x0010
0x0010 << 4 == 0x0100
0x0100 << 4 == 0x1000

10.3 Five Governors

FACTS	p = -1
RULES	p = 0
CLOSURES	p = 1

```
COMBINATORS  p = 2
CONS         p = 3
```

Governor order:

```
The exponents are consecutive from -1 through 3.
RULES is the p=0 Genesis/equality pivot.
FACTS and CONS are circular inverse endpoints.
```

Readable form:

```
FACTS ground.
RULES permit.
CLOSURES seal.
COMBINATORS compose.
CONS embodies.
```

Or in older OMI canon:

```
Rules declare.
Facts ground.
Closures seal.
Combinators compose.
CONS reduces.
```

11. Miquel Incidence and the 11-Cell Oversight Frame

The Tetragrammatron is not merely four gauges.

It is the Miquel incidence controller generated by four gauges under open/sealed polarity.

Four gauges:

```
FS
GS
RS
US
```

Two boundary states:

sealed
open

This gives eight points:

sealed FS
sealed GS
sealed RS
sealed US
open FS
open GS
open RS
open US

The six pairwise gauge planes are:

FS-GS
FS-RS
FS-US
GS-RS
GS-US
RS-US

Miquel incidence counts:

points = 8
circles = 6
point degree = 3
circle degree = 4
flags = 24

Balance:

$$8 * 3 = 6 * 4 = 24$$

The oversight arithmetic is:

hidden 5-cell = 5
pairwise gauge planes = $C(4,2) = 6$
oversight frame = $5 + 6 = 11$

Canonical sentence:

5-cell hides.
6 gauge-pair planes oversee.
11-cell governs.
Projection lies.
Receipt accepts.

Boundary:

The $5 + 6 = 11$ frame is an OMI oversight arithmetic.
It is not automatically identified with the abstract regular 11-cell without an incidence-preserving proof.

12. Polybius Square as Invariant Set Logic

The Polybius square:

0	1	2	3
4	5	6	7
8	9	A	B
C	D	E	F

Primary diagonal:

$D^+ = \{0, 5, A, F\}$

Binary:

0000
0101
1010
1111

XOR:

$0000 \wedge 0101 \wedge 1010 \wedge 1111 = 0000$

Anti-diagonal:

$$D^- = \{3, 6, 9, C\}$$

Binary:

```
0011
0110
1001
1100
```

XOR:

$$0011 \wedge 0110 \wedge 1001 \wedge 1100 = 0000$$

Key observation:

Each diagonal XORs to zero.
This is exact, byte-wide, platform-independent closure.
No floating point is involved.

This is non-destructive closure.

The wheel can be inspected at any stage.

The structure proves itself through exact arithmetic rather than by collapsing into a single witness bit.

13. Binary Quadratic Form

The OMI binary quadratic form is:

$$Q(x,y) = 60x^2 + 16xy + 4y^2$$

It separates the three major planes:

$4y^2$	Atomic Logic Clock
$16xy$	Spectral Observer / Bridge Plane
$60x^2$	Cosmic Orbit Clock

Important arithmetic correction:

$$\begin{aligned} 60x^2 + 16xy + 4y^2 \\ = 4(15x^2 + 4xy + y^2) \end{aligned}$$

It does not factor as:

$$4(3x + y)(5x + y)$$

because:

$$\begin{aligned} 4(3x + y)(5x + y) \\ = 4(15x^2 + 8xy + y^2) \\ = 60x^2 + 32xy + 4y^2 \end{aligned}$$

The discriminant of the original form is:

$$\begin{aligned} b^2 - 4ac &= 16^2 - 4(60)(4) \\ &= 256 - 960 \\ &= -704 \end{aligned}$$

Since:

$$\begin{aligned} a &= 60 > 0 \\ \text{discriminant} &< 0 \end{aligned}$$

the form is positive definite over the real plane.

Correct statement:

$Q(x,y)$ is positive definite.
It has no real linear factorization.
Its roots live in a complex/projective interpretation, not as real transition lines.

Boundary:

The quadratic form organizes OMI plane weights.
It does not by itself prove a physical geometry.

14. OMI-ISA Witness Split

The executable witness fixtures should remain separated:

NULL Ring:
0x00 / 0x20 / 0x7F / 0xFF
byte-ring closure

Tetragrammatron:
0x1E / 0x3C / 0x78
diagonal closure, fold-carry, polyharmonic governor

Metatron:
0xAA55 / 0x55AA
alternating balanced witness

Canonical lock:

0x3C / 0x78 belongs to Tetragrammatron.
0xAA55 / 0x55AA belongs to Metatron.
0x00 / 0x20 / 0x7F / 0xFF belongs to the Null Ring.

15. Proof Boundary

The proof rail belongs to Coq.

The executable rail belongs to OMI-ISA.

The notation rail belongs to OMI workbook / OMI-Port.

Correct propagation:

Coq proves.
Canon names.
OMI workbook notation cites.
OMI-Port routes.

```
OMI-Lisp may declare.  
OMI-ISA witnesses.  
Tetragrammatron validates.  
Metatron scribes.  
Receipt records.
```

C tests and OMI-ISA fixtures do not prove the mathematics.

They witness bounded executable behavior aligned with already-proven or proof-scaffolded invariants.

Boundary:

```
No projection is authority.  
No route is connection.  
No notation is receipt.  
No gauge is truth.
```

16. Final Canon

The hidden control plane establishes place.

FS, GS, RS, and US close the pre-language gauge control line.

Space at 0x20 opens readable separation.

The F-column branches the hidden unit separator into printable notation:

US / ? 0 _ o

The Null Ring closes the byte cycle:

0x00 → 0x20 → 0x7F → 0xFF → 0x00

The Tetragrammatron adjudicates diagonal closure through:

0x1E, 0x3C, and 0x78

The Metatron scribes contrast through:

0xAA55 and 0x55AA

Omicron frames the resolver envelope:

F8 00 1C 1D 1E 1F 20 F8

OMI notation cites relation as:

omi---imo?O_o

Receipt alone records accepted closure.

Shortest form:

US hides.

Slash separates.

Question witnesses.

O norms.

Underscore carries.

o localizes.

Null Ring closes.

Tetragrammatron validates.

Metatron scribes.

Omicron frames.

Receipt records.